LIN7209 – Syntax

Cartography

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Queen Mary University of London

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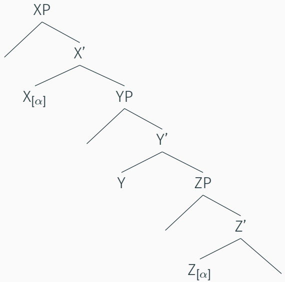

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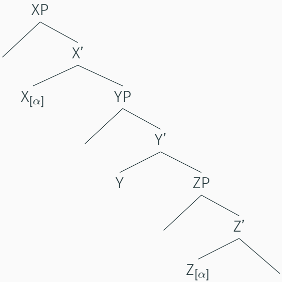

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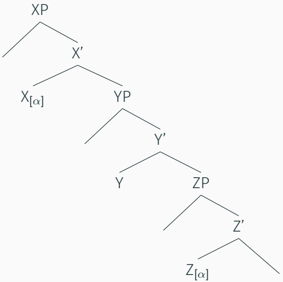

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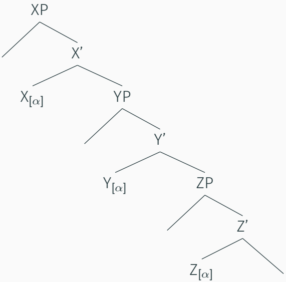

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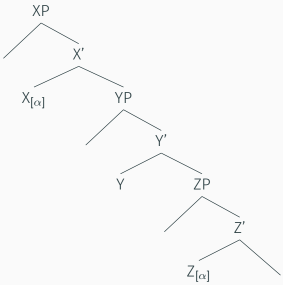

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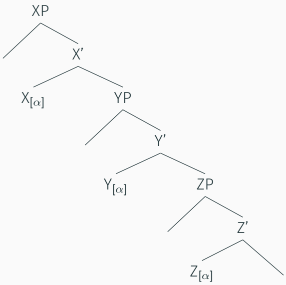

Agreement between X and Z violates Relativized Minimality

The nature and locus of functional heads

- Functional heads tend to be "high", trigger **movement**to their Spec.
- In Minimalism, they are responsible **for many surface phenomena**: movement, morphological agreement/inflection.
- Heads correspond to "**criterial**" positions. For Rizzi, these have a semantic or pragmatic function.
- What are they exactly, and were are they located?

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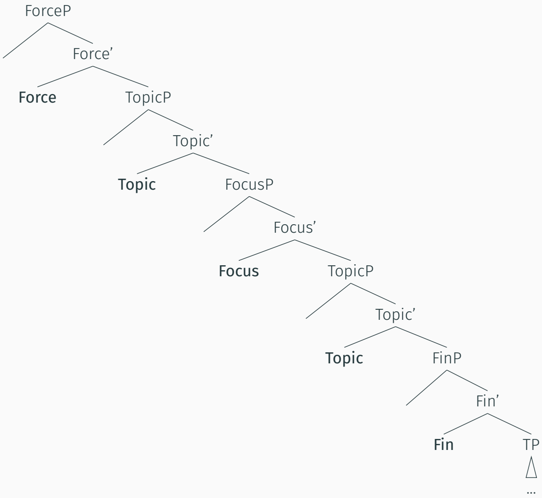

Cartography in different domains

- In the Cinque and Rizzi paper, this same idea is applied to the other “domains” of the clause and DP.
- The claim is that they these domains are equally as complex, and **one can draw “maps” of them**, charting the different functional categories.

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The CP domain

Italian complementizers

- In Italian, there are **two different complementizers** depending on whether the embedded clause is finite (and declarative), or infinitival.
  - For finite declarative clauses, *che* will be used.
  - For infinitival (“control”) clauses, the preposition *di* get used.

(1) Credo che loro apprezzerebbero molto il tuo libro.
I-believe CHE they appreciate.COND a-lot the your book.
‘I believe that they would appreciate your book very much.

(2) Credo di apprezzare molto il tuo libro.
I-believe DI appreciate.INF a-lot the your book.
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Credo	che	loro	apprezzerebbero	molto	il	tuo	libro.
I-believe	CHE	they	appreciate.COND	a-lot	the	your	book.

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Italian complementizers and Topic

- Does having different complementizers tell us that we need two different C-heads? Not quite: a single head could be realized in two different ways on the surface!
- Idea: assess whether *che* and *di* **linearize differently w.r.t. another left-peripheral element**.
- If so, this is evidence that *che* and *di* are hosted in distinct head, a higher one and a lower one, divided by a third projection.

Che and embedded topics

(3)

Credo	che	il	tuo	libro,	loro	lo	apprezzerebbero	molto.
I-believe	CHE	the	your	book	they	it	appreciate.COND	a-lot.

‘I believe that your book, they would appreciate it very much.’

(4)

* Credo,	il	tuo	libro,	che	loro	lo	apprezzerebbero	molto.
I-believe	the	your	book	CHE	they	it	appreciate.COND	a-lot.

Intended: ‘I believe that your book, they would appreciate it very much.’

Che and embedded topics

(3) Credo che il tuo libro, loro lo apprezzerebbero molto.
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- *Che* must **precede** a topicalized DP.

(5) * Credo di il tuo libro, apprezzarlo molto.
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Intended: ‘Your book, I believe to appreciate it very much.’

(6) Credo il tuo libro, di apprezzarlo molto.
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Di and embedded topics

(5) * Credo di il tuo libro, apprezzarlo molto.

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- *Di* must **follow** a topicalized DP.

Two different C-heads with slightly different functions

- *Che* must precede an embedded topic, while *di* must follow it.
- Conclusion: Italian complementizers can appear both "up" and "down".
- This is evidence that complementizers can be hosted **by two different functional heads**.
- In Italian, they'll take different surface forms.

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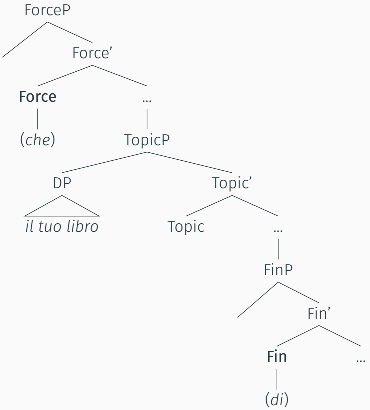
Force and Fin(iteness)

- *Che* occupies a higher **Force** head, which determines clause type (declarative, question, exclamative…).
- *Di* occupies a lower Finiteness (**Fin**) head.
- When tense/mood distinctions are made (finite case), they come along with richer information in general; while non-finite clauses tend not to have nominatives, less agreement etc.).



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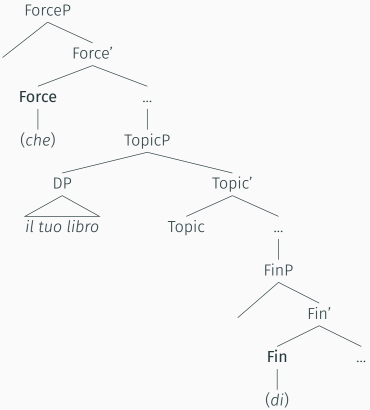

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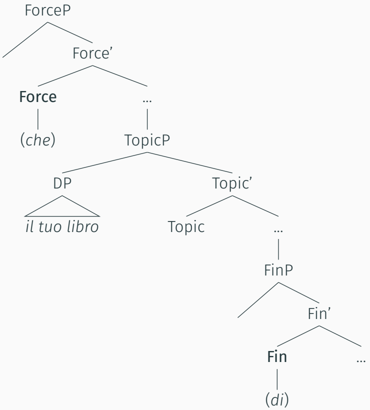

Extra evidence from Welsh

- We've just seen that different complementizers interact differently with other left-peripheral elements like topics.

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(7) Dywedais i mai fel arfer y dynion a fuasai'n gwerthu'r ci.
Said I MAI as usual the men PRT would Asp sell-the dog.
'I said that it's as usual the men who would sell the dog.'

- This also shows the variety of things that can be "sandwiched" in between: here a focus and a preposed adverbial.

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The TP/IP domain

Mapping the IP domain

- So far we have been assuming modals, auxiliaries, tense... lived in the same TP projection.
- However, these elements appear to **fulfill different functions**, so it's not crazy to assume that they should be represented as **heads of distinct projections**.
- Extra evidence for this again comes from **relative ordering**: if tense, modals, mood, aspect... lived in the same head, we'd expect them to be linearized in arbitrary orders cross-linguistically.
- But the fact is that **these elements seem to follow a strict hierarchical ordering cross-linguistically**. We'll see partial evidence for this, focusing on pairs of such elements: (Mood, Tense) and (Tense, Aspect).

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Ordering of epistemic must (Mood) and Tense

- Epistemic mood morphemes appear further away from the verb than tense. From Thai (Cinque, 1999):

(8)

lom	khong	cà?	kamlang	phát
wind	EPIST	FUT	PROG	blow

'The wind must be blowing.'

- This is also the case in languages in which these morphemes follow the verb. From Garo (Sino-Tibetan) (Cinque, 1999):

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Anti-ci	re'an-aha-kon
market-to	go-PAST-EPIST

'He must have gone to the market.'

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i	pahn	kin	kangkang	rais
I	FUT	HAB	eat.PROG	rice

'I will habitually be eating rice.'

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Nunqan	ulle-ve	tulile	lo:van-d'e-ngne-re-n
she	meat-ACC	outdoors	hang-IMPF-HAB-PAST-3sg

'She used to hang meat outdoors for some time (for drying).'

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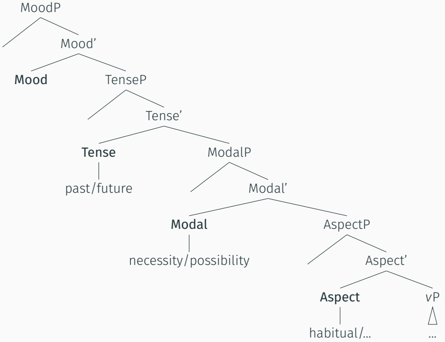
Simplified cartography of the IP domain



- The full "map" given in the Rizzi and Cinque paper has some specific tense and modal heads showing up downstairs in the "Aspect" area. But roughly, it looks like the one upstairs.

Adverb placement mirrors the IP cartography

- **Adverbs** can be more mood-y, tense-y, modal-y, or aspect-y.
- From the Rizzi and Cinque paper: *frankly* is related to Mood, *probably* to Modality (possibility), *usually* and *frequently* to Aspect (habitual/frequentative aspect).
- Looking at their ordering constitutes additional evidence that Mood, Tense, Modality, Aspect, are rigidly and hierarchically ordered. It can also help **further break down these domains**.

(12) a. (Frankly), Jo will (frankly) probably (*frankly) show up.
Mood > Modal

b. (Frankly), Jo (frankly) usually (*frankly) shows up.
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Adverb placement cross-linguistically

- The ordering facts are robust cross-linguistically.
- The Rizzi and Cinque paper also mentions the ordering of various kinds of PPs in German and Japanese.
- A radical, semantic alternative (Scontras et al., 2017; Hahn et al., 2018): it's not about different syntactic heads, it's all about **subjectivity**!
- Under that view, *frankly* (Mood) > *probably* (Modality) > usually (Aspect) > *frequently* (Aspect) are ordered on a scale of subjectivity.
- **Less subjective** elements (whether they are adjectives, adverbs...) are **closer** to what they modify, while **more subjective** ones are **further away**.
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- **Less subjective** elements (whether they are adjectives, adverbs...) are **closer** to what they modify, while **more subjective** ones are **further away**.
- Rationale: this strategy minimizes potential misunderstandings, and has nothing to do with syntax!

Adverb placement cross-linguistically

- The ordering facts are robust cross-linguistically.
- The Rizzi and Cinque paper also mentions the ordering of various kinds of PPs in German and Japanese.
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The DP domain

Determiners, Numerals, Adjectives, and Nouns

- Initial evidence that the DP domain has a "cartography" too: stable hierarchical ordering of determiners, numerals, adjectives, and nouns, again regardless of linear order.

- Orderings in line with this hierarchical structure: Det-Num-A-N, Det-Num-N-A, Det-A-N-Num, Det-N-A-Num, Num-A-N-Det, Num-N-A-Det, A-N-Num-Det, N-A-Num-Det, i.e. 8 out of 24 theoretically possible orderings in total. Some more common than others.

- Experimental evidence (Culbertson & Adger, 2014): language learners, when confronted with a new language, privilege structural similarity to their native language over surface statistical similarity.

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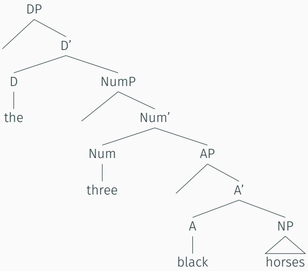

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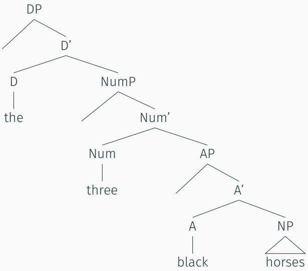

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The adjectival layer

- Subsequent research expanded the adjective-related functional projections to encompass at least the following classes: subjective comment > evidential > size > length > height > speed > depth > width > weight > temperature > wetness > age > shape > color > nationality/origin > material > compound element > NP.
- Again debate on whether this should be syntactically encoded, or is simply about subjectivity and optimal information transfer.

Adjacency effects

Comp-trace effects in English

(13) Jo said Al saw Ed.

(14) Who did Jo say (that) Al saw ~~Ed~~?

(15) Who did Jo say (*that) ~~Al~~ saw Ed?

- When an embedded **object** is *wh*-extracted, an overt complementizer may be present.
- When an embedded **subject** is *wh*-extracted, no overt complementizer can be present.
- Empirical reformulation: overt complementizers do not like to be close to *wh*-traces!

Comp-trace effects in Nupe

- From Kandybowicz (2008):

(16)

Ke	u:	bè	[ke	Musa	du	] na	o?
what	3.SG	seem	COMP	Musa	cook	NA	O

'What does it seem that Musa cooked?'

(17)

* Zèé	u:	bè	[ke	du	nakàn]	na	o?
who	3.SG	seem	COMP	cook	meat	NA	O

Intended: 'Who does it seem cooked meat?'

- Nupe is similar to English.

Comp-trace effects in French

- Original observation by Perlmutter (1971);[2]

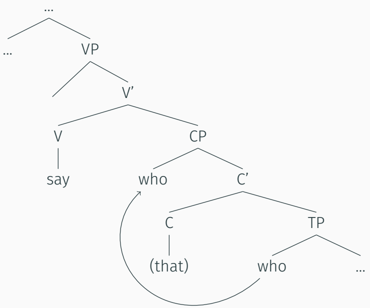
(18) Qui a-t-il dit que Marie voulait voir?
who has-he said that Marie wanted to see
'Who did he say that Marie wanted to see?'

(19) Qui a-t-il dit qui voulait voir Marie?
who has-he said qui wanted to.see Marie
'Who did he say wanted to see Marie?'

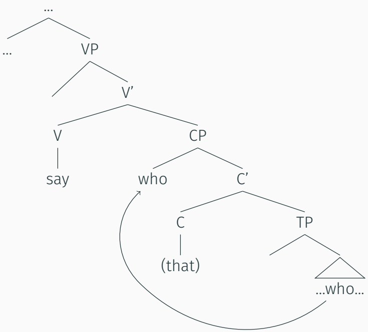
- In case of object *wh*-extraction, we get the standard complementizer *que*.
- In case of subject *wh*-extraction, we get *qui*.
- Having a trace near a complementizer does not force its deletion, but instead alters it!

[2] I'm not crazy about this example, but I get the effect for other similar examples with different verbs. See also Bùlì for related effects (Hiraiwa, 2003).

Sketch of an analysis



Subject *wh*-extraction



Object *wh*-extraction

- With subject extraction, movement is local, from Spec-TP to Spec-CP. C does not like that, and gets silent/altered.
- With object extraction, movement is less local, from inside VP to Spec-CP. C is fine with this.
- Specific implementation: head government; antilocality constraint...

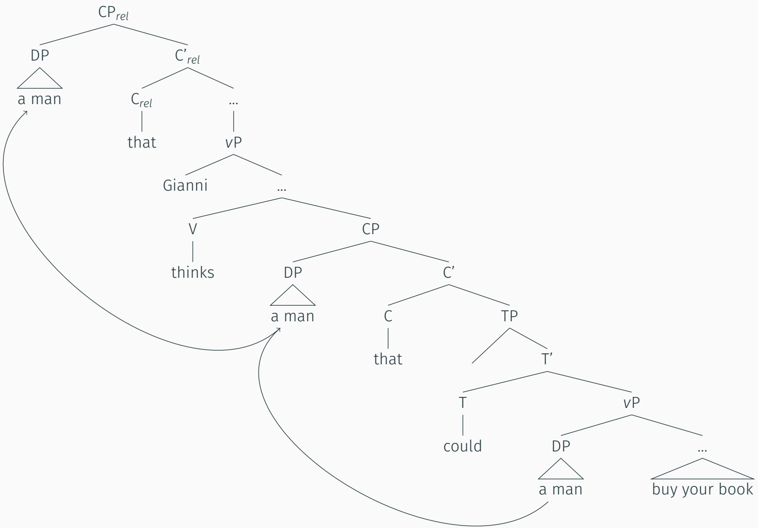
An interesting prediction for Italian

- If Comp-trace effects indeed occur when movement is too local, from a Spec to the Spec immediately above, then languages in which subject movement need not proceed via Spec-TP should be devoid of Comp-trace effects.
- Recall Italian sentences to not need an overt subject, i.e. Spec-TP maybe empty. What do we predict then?

(20) un uomo che Gianni pensa que potrebbe comprare il tuo libro
a man that Gianni thinks that could buy the your book
'A man that Gianni thinks could buy your book.'

- Despite the fact we relativized an embedded subject, an overt complementizer is licensed!

Italian subject *wh*-extraction



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